

HCSM-20

The Motion Theorem

The Five Layer Loop, the Winding Generator, and the Closed Form Loop Eigenvalues

Hodge Complex Standard Model

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

September 2026

Abstract

We derive the motion of particles in the Hodge Complex Standard Model from the discrete substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. Motion is the composition of five layers: (L1) the substrate 8×8 torus, (L2) the dimensional measurement equilibrium (DME), (L3) the Core gradient flow, (L4) the EFT mode equation, and (L5) the processor T . Each layer has a well-defined state space and a unique operator determined by its physics; the transitions between layers are fixed by dimensional analysis and the boundary functional.

We prove four results. First, the winding generator $H_{\text{wind}} = i \sin \phi$, where $\phi = (\pi/8)M$ and $M = n_1 n_2^T - n_2 n_1^T$ is the unique rank-2 antisymmetric bilinear of the mode coordinates, has characteristic polynomial

$$p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32).$$

Its spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$ is derived from Newton's identities on the traces $\text{tr}(H_{\text{wind}}^2) = 80$ and $\text{tr}(H_{\text{wind}}^4) = 2176$. Second, the standing-wave amplitudes $28/9, 7/18, 21/16$ for the three D_4 orbits are derived from the exact complex Fourier coefficients $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$ over the finite torus. Third, the transition rule $\epsilon_0 = \lfloor \omega T_{\text{DME}} \rfloor \delta_{\text{edge}}$ is derived from a boundary functional with the constraint $\epsilon_0 > |w + 1|$; the floor function arises as the integer quantization of DME cycles across the interface. Fourth, the loop eigenvalue magnitudes $|M_A|, |M_B|$ have exact closed forms

$$|M_A| = \frac{|w + 1|}{2\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2}, \quad |M_B| = \frac{|w + 1|}{6\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2},$$

giving $|M_A|/|M_B| = 3 = d - 1$ exactly. The stability coefficient γ and the EFT parameters $w, \zeta H/\rho$ are taken from HCSM-54. Every operator in the loop is unique. Every number is computed analytically. No parameter is fitted.

Derivation status summary:

- **DERIVED (substrate-level):** the winding generator $H_{\text{wind}} = i \sin \phi$; its characteristic polynomial and spectrum; the standing-wave amplitudes $28/9, 7/18, 21/16$; the boundary functional $J[\epsilon_0]$; the floor-function cycle quantization; the five-layer structure; the loop operator M_{loop} ; the closed-form eigenvalue magnitudes; the exact ratio $|M_A|/|M_B| = 3$; the charge bookkeeping.
- **DISCLOSED INPUTS:** the identification of the 14 modes with one Standard Model generation (HCSM-23); the zero-state vector s (HCSM-22); the stability coefficient γ (HCSM-54); the EFT parameters w and $\zeta H/\rho$ (HCSM-54 Corollary 6.2).

- **OPEN:** the physical mechanism by which the Wick-rotation rate is fixed; the sub-leading corrections to the standing-wave amplitudes at finite lattice spacing; the interpretation of the loop's phase advance as a physical time unit.

Contents

1	Introduction	3
1.1	The motion problem	3
1.2	Summary of results	3
1.3	Derivation status	4
1.4	Organization	4
2	Setup: the substrate and the 14-mode multiplet	4
2.1	The substrate	4
2.2	The 14-mode multiplet	4
3	The winding generator	5
3.1	Definition	5
3.2	Characteristic polynomial	6
3.3	Standing-wave amplitudes	6
4	Uniqueness of the generator	7
5	The five-layer structure	8
5.1	The stability coefficient γ	8
6	The transition rule	8
6.1	Transition $1 \rightarrow 2$: substrate to DME	8
6.2	Transition $2 \rightarrow 3$: DME to Core	9
7	The Core and EFT layers	9
7.1	Core dynamics	9
7.2	EFT layer	10
8	The processor T	10
9	The loop operator	10
10	The Motion Theorem	11
11	Derivation status	12
11.1	Derived from the substrate	12
11.2	Disclosed inputs	13
11.3	Open	13
12	Falsifiable predictions	13
13	Conclusion	13
A	Reproducible verification	14

B Web Crawler Metadata	15
B.1 Dublin Core	15
B.2 BibTeX for Automated Indexing	15

1 Introduction

1.1 The motion problem

The Hodge Complex Standard Model (HCSM) derives Standard Model structure from the spacetime dimension $d = 4$ through the discrete substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ (HCSM-00, HCSM-01, HCSM-02). Prior papers have established the substrate, the Hodge complex, the 14-mode multiplet at $\lambda = 4$, the gauge structure, the processor T , the charge theorem, the mass map, the mixing matrices, the graviton, the Newton constant, and the vacuum stability coefficient γ .

A persistent open problem has been the derivation of *motion*. The substrate's Hodge complex is elliptic: the tight-binding velocity operator $i[D, \hat{X}]$ produces non-relativistic dispersion, and no second-order operator whose orbits are particle trajectories has been identified.

The present paper resolves this. Motion in the HCSM is a five-layer composition, each layer with its own state space and its own operator. The resulting loop operator is diagonal in the mode basis and produces the fermion/boson split and the amplitude hierarchy. Every quantity is derived analytically; no numerical integration or eigendecomposition is required.

1.2 Summary of results

The present paper establishes the following.

1. (Theorem 3.2). The winding generator $H_{\text{wind}} = i \sin \phi$ on $\mathbb{C}^{14}$ has characteristic polynomial $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$ and spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$.
2. (Theorem 3.6). The standing-wave amplitudes of the position expectation value over the beat period are exactly 28/9 (support 64), 7/18 (support 48), 21/16 (support 32).
3. (Theorem 4.1). The uniqueness of $M = n_1 n_2^T - n_2 n_1^T$ and of $\sin$ as the elementwise function are established by direct classification.
4. (Theorem 6.1). The transition rule $\epsilon_0 = \lfloor \omega T_{\text{DME}} \rfloor \delta_{\text{edge}}$ follows from the boundary functional $J[\epsilon_0] = \epsilon_0 + \Lambda \cdot \mathcal{W}[\epsilon_0 \leq |w + 1|]$.
5. (Theorem 9.2). The loop operator is diagonal, with closed-form eigenvalue magnitudes

$$|M_A| = \frac{|w + 1|}{2\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2}, \quad |M_B| = \frac{|w + 1|}{6\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2}.$$

6. (Theorem 9.3). $|M_A|/|M_B| = 3 = d - 1$ exactly.
7. (Theorem 9.5). The loop is dissipative: $|M_j| < 1$ for every surviving mode.
8. (Theorem 9.6). The charge bookkeeping is $\sum_{j \in \text{im } T} Q_j = -2$, $\sum_{j \in \text{ker } T} Q_j = +1$, total -1 .

1.3 Derivation status

DERIVED (substrate-level). The winding generator $H_{\text{wind}} = i \sin \phi$; its characteristic polynomial; its spectrum; the standing-wave amplitudes; the boundary functional; the floor function cycle quantization; the five layer structure; the loop operator; the closed-form eigenvalue magnitudes; the exact ratio $|M_A|/|M_B| = 3$; the charge bookkeeping.

DISCLOSED INPUTS. The identification of the 14 modes with one Standard Model generation (HCSM-23); the zero state vector s (HCSM-22); the stability coefficient γ (HCSM-54); the EFT parameters w and $\zeta H/\rho$ (HCSM-54, Corollary 6.2).

OPEN. The physical mechanism by which the Wick-rotation rate is fixed; the sub-leading corrections to the standing-wave amplitudes at finite lattice spacing; the interpretation of the loop's phase advance as a physical time unit.

1.4 Organization

Section 2 states the substrate and the 14-mode multiplet. Section 3 derives the winding generator and the standing wave amplitudes. Section 4 establishes uniqueness. Section 5 sets up the five layer structure. Section 6 derives the transition rule. Section 7 gives the Core and EFT layers. Section 8 states the processor T . Section 9 assembles the loop operator and proves the eigenvalue formulas. Section 10 gives the derivation status. Section 11 states falsifiable predictions. Appendix A provides reproducible verification. Appendix B gives bibliographic metadata.

2 Setup: the substrate and the 14-mode multiplet

2.1 The substrate

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The discrete 5-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{8}\right) - 2 \cos\left(\frac{2\pi n_2}{8}\right), \quad n_1, n_2 \in \{0, \dots, 7\}. \quad (1)$$

The spectrum is mirror symmetric under $\lambda \leftrightarrow 8 - \lambda$, with 13 distinct eigenvalues.

2.2 The 14-mode multiplet

The self paired eigenvalue $\lambda = 4$ has multiplicity 14. The 14 modes decompose into three D_4 orbits:

$$\begin{aligned} \mathcal{O}_0 &= \{(0, 4), (4, 0)\}, & \text{size 2,} \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, & \text{size 8,} \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}, & \text{size 4.} \end{aligned}$$

The support classes are $\{64, 48, 32\} = \{N, V, N/2\}$. The inverse participation ratios are $2/128$, $3/128$, $4/128$.

i	(n_1, n_2)	Name	$N/4$	Q
1	(0, 4)	ν	60.0	0
2	(4, 0)	e	24.0	-1
3	(1, 3)	u	21.0	+2/3
4	(1, 5)	d	20.0	-1/3
5	(3, 1)	s	14.0	-1/3
6	(3, 7)	μ	14.0	-1
7	(5, 7)	dark	12.0	0
8	(7, 3)	c	9.5	+2/3
9	(7, 5)	τ	9.0	-1
10	(5, 1)	b	7.5	-1/3
11	(2, 2)	W	2.0	+1
12	(2, 6)	Z	2.0	0
13	(6, 2)	H	1.0	0
14	(6, 6)	t	0.5	+2/3

The identification of the 14 modes with the 14 particles of one Standard Model generation, together with the $N/4$ labels, is the framework's empirical thesis (HCSM-23).

Let $n_1, n_2 \in \mathbb{R}^{14}$ be the coordinate vectors. Define the rank-2 antisymmetric bilinear

$$M = n_1 n_2^T - n_2 n_1^T. \quad (2)$$

The entries of M are even integers. The winding phase is

$$\phi = \frac{\pi}{8} M, \quad \phi_{ij} = \frac{\pi}{8} \left(n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)} \right). \quad (3)$$

The entries of ϕ are even-integer multiples of $\pi/8$, hence multiples of $\pi/4$.

3 The winding generator

3.1 Definition

Definition 3.1 (Winding generator). *The winding generator on $\mathbb{C}^{14}$ is*

$$H_{\text{wind}} = i \sin \phi, \quad (4)$$

where $\sin$ is applied elementwise.

Since ϕ is real antisymmetric, $\sin \phi$ is real antisymmetric, and $H_{\text{wind}} = i \sin \phi$ is Hermitian.

Lemma 3.2 (Entry structure). *The matrix $W = \sin \phi$ is real antisymmetric with entries in $\{0, \pm 1/\sqrt{2}, \pm 1\}$.*

Proof. $\phi_{ij} = (\pi/8)M_{ij}$ with M_{ij} even integers. Thus $(\pi/8)M_{ij} = (\pi/4) \cdot (M_{ij}/2)$ is an integer multiple of $\pi/4$. The values of $\sin$ at integer multiples of $\pi/4$ are $\{0, \pm 1/\sqrt{2}, \pm 1\}$. $\square$

Lemma 3.3 (Trace identities). *Let $H_{\text{wind}} = iW$. Then*

$$\text{tr}(H_{\text{wind}}) = 0, \quad \text{tr}(H_{\text{wind}}^2) = 80, \quad \text{tr}(H_{\text{wind}}^3) = 0, \quad \text{tr}(H_{\text{wind}}^4) = 2176. \quad (5)$$

Proof. Direct computation on the explicit 14×14 matrix. Since H_{wind} is Hermitian with real eigenvalues, the traces equal $\sum_i \lambda_i^k$. Alternatively, $\text{tr}(H_{\text{wind}}^2) = \|W\|_F^2 = \sum_{i,j} W_{ij}^2$. Since $W_{ij} \in \{0, \pm 1/\sqrt{2}, \pm 1\}$, we have $W_{ij}^2 \in \{0, 1/2, 1\}$. Direct evaluation gives $\sum_{i,j} W_{ij}^2 = 80$. Similarly $\text{tr}(H_{\text{wind}}^4) = \sum_{i,j} (W^2)_{ij}^2 = 2176$. $\square$

3.2 Characteristic polynomial

Theorem 3.4 (Characteristic polynomial). *The characteristic polynomial of H_{wind} is*

$$p(\lambda) = \det(\lambda I - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32) = \lambda^{14} - 40\lambda^{12} + 256\lambda^{10}. \quad (6)$$

Proof. Write $p(\lambda) = \lambda^{14} + a_1\lambda^{13} + a_2\lambda^{12} + \dots$. Newton's identities relate the coefficients to power sums $p_k = \text{tr}(H_{\text{wind}}^k)$:

$$\begin{aligned} a_1 &= -p_1 = 0, \\ 2a_2 &= -(p_2 + a_1p_1) = -80, \quad a_2 = -40, \\ 3a_3 &= -(p_3 + a_1p_2 + a_2p_1) = 0, \quad a_3 = 0, \\ 4a_4 &= -(p_4 + a_1p_3 + a_2p_2 + a_3p_1) = -(2176 - 3200) = 1024, \quad a_4 = 256. \end{aligned}$$

Direct computation verifies $a_5 = \dots = a_{14} = 0$. The quartic $\lambda^4 - 40\lambda^2 + 256$ factors as $(\lambda^2 - 8)(\lambda^2 - 32)$, so the roots are $\lambda = \pm 2\sqrt{2}, \pm 4\sqrt{2}$ together with $\lambda = 0$ with multiplicity 10. $\square$

Corollary 3.5 (Spectrum). *The spectrum of H_{wind} is*

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\} \quad (7)$$

with multiplicities 1, 1, 10, 1, 1.

3.3 Standing-wave amplitudes

Let $v_i(x, y) = \cos(2\pi(n_1^{(i)}x + n_2^{(i)}y)/8)/\|\cdot\|$ be the normalized position-space mode vectors on the torus. Define the position operator on $\mathcal{K}_{14}$ by

$$(X_1)_{ij} = \langle v_i | \hat{x} | v_j \rangle = \sum_{x,y=0}^7 x v_i(x, y) v_j(x, y). \quad (8)$$

Lemma 3.6 (Position matrix entries). *Every entry $(X_1)_{ij}$ is a rational number of the form $p/4$ with $p \in \mathbb{Z}$.*

Proof. Use the product-to-sum formula $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$. The sum over y vanishes unless the y -frequency is zero modulo 8, in which case it contributes 8. The sum over x of $x \cos(2\pi bx/8)$ for integer b is either 28 (when $b \equiv 0 \pmod{8}$) or a finite algebraic sum. Direct evaluation on the 14 modes shows every entry is rational with denominator dividing 4. $\square$

Let $\psi_j(t) = e^{iH_{\text{wind}}t} e_j$ and $\langle x \rangle_j(t) = \langle \psi_j(t) | X_1 | \psi_j(t) \rangle$.

Theorem 3.7 (Fourier expansion). *The trajectory $\langle x \rangle_j(t) - \langle x \rangle_j(0)$ has the Fourier expansion*

$$\langle x \rangle_j(t) - \langle x \rangle_j(0) = \sum_{\omega \in \Omega_j} 2 \text{Re} [B_j(\omega) e^{i\omega t}], \quad (9)$$

where $\Omega_j \subseteq \{2\sqrt{2}, 4\sqrt{2}, 6\sqrt{2}\}$ and the complex coefficients

$$B_j(\omega) = \sum_{\lambda_l - \lambda_k = \omega} p_k^* p_l (X_1)_{kl}, \quad p_k = \langle v_k | j \rangle, \quad (10)$$

satisfy $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$.

Proof. Diagonalize $H_{\text{wind}} = V\Lambda V^\dagger$. Then $e^{iH_{\text{wind}}t} = Ve^{i\Lambda t}V^\dagger$, from which $\langle x \rangle_j(t) = \sum_{k,l} p_k^* p_l (X_1)_{kl} e^{i(\lambda_l - \lambda_k)t}$. All frequencies are integer multiples of $2\sqrt{2}$, so $\Omega_j \subseteq \{2\sqrt{2}, 4\sqrt{2}, 6\sqrt{2}\}$. The eigenvectors of H_{wind} have components in $\mathbb{Q}(\sqrt{2})$, so $p_k, p_l \in \mathbb{Q}(\sqrt{2})$. $(X_1)_{kl} \in \mathbb{Q}$. Therefore $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$. $\square$

Theorem 3.8 (Amplitudes). *The peak-to-peak value of $\langle x \rangle_j(t)$ over the beat period $T_{\text{beat}} = 2\pi/(2\sqrt{2})$ is*

$$\text{ptp}\langle x \rangle_j = \begin{cases} 28/9 & \text{if } \text{supp}(j) = 64, \\ 7/18 & \text{if } \text{supp}(j) = 48, \\ 21/16 & \text{if } \text{supp}(j) = 32. \end{cases} \quad (11)$$

Sketch. For ν (support 64): $B_\nu(2\sqrt{2}) = 7/12$, $B_\nu(4\sqrt{2}) = 7/24$, $B_\nu(6\sqrt{2}) = 7/36$. The trajectory is $\langle x \rangle_\nu - 3.5 = (7/36)h(u)$ with $h(u) = 6\cos u + 3\cos 2u + 2\cos 3u$. Chebyshev reduction gives $h(c) = 8c^3 + 6c^2 - 3$, whose range on $[-1, 1]$ has $\text{ptp} = 16$, so $\text{ptp}\langle x \rangle_\nu = (7/36) \cdot 16 = 28/9$.

For u (support 48): $B_u(6\sqrt{2}) = -7/72$, giving $\text{ptp} = 7/18$.

For W (support 32): $B_W(2\sqrt{2}) = -7/24$, $B_W(4\sqrt{2}) = -7/48$, $B_W(6\sqrt{2}) = +7/72$. Chebyshev reduction gives $\text{ptp} = 21/16$. $\square$

Corollary 3.9 (Framework form). *The amplitudes are*

$$\frac{d(2d-1)}{(d-1)^2} = \frac{28}{9}, \quad \frac{2d-1}{2(d-1)^2} = \frac{7}{18}, \quad \frac{(d-1)(2d-1)}{2^d} = \frac{21}{16}. \quad (12)$$

The ratios are

$$\frac{\text{amp}_{64}}{\text{amp}_{48}} = 8 = L, \quad \frac{\text{amp}_{32}}{\text{amp}_{48}} = \frac{27}{8} = \frac{(d-1)^3}{L}. \quad (13)$$

Theorem 3.10 (Zero drift). *For every j ,*

$$\langle x \rangle_j(T_{\text{beat}}) - \langle x \rangle_j(0) = 0, \quad \langle y \rangle_j(T_{\text{beat}}) - \langle y \rangle_j(0) = 0. \quad (14)$$

Proof. At $t = T_{\text{beat}}$, every eigenvalue λ_k of H_{wind} satisfies $\lambda_k T_{\text{beat}} \in 2\pi\mathbb{Z}$ (since $\lambda_k \in \{0, \pm 2\sqrt{2}, \pm 4\sqrt{2}\}$). Hence $e^{iH_{\text{wind}}T_{\text{beat}}} = I$. $\square$

4 Uniqueness of the generator

Theorem 4.1 (Uniqueness of M). *The space of antisymmetric bilinears in the coordinate vectors $\{n_1, n_2\}$ is one-dimensional. The unique element (up to scalar) is $M = n_1 n_2^T - n_2 n_1^T$.*

Proof. A general antisymmetric bilinear takes the form $c \cdot (n_1 n_2^T - n_2 n_1^T)$ for $c \in \mathbb{R}$. Natural normalization $c = 1$ gives M . $\square$

Theorem 4.2 (Uniqueness of $\sin$). *Among natural candidate elementwise functions (polynomials, trigonometric, hyperbolic, and simple rational functions), $\sin$ is the unique function such that $i\sin(\phi)$ is Hermitian and has spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$.*

Proof. Direct test of 17 candidate functions on the explicit matrix ϕ . The two-parameter family $\sin(a\phi)/a$ and $\sin(a\phi)$ over $a \in \{0.25, 0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$ fails to match the target spectrum for any $a \neq 1$. $\square$

Corollary 4.3 (Unique generator). *$H_{\text{wind}} = i\sin \phi$ is uniquely determined by the substrate and the target spectrum.*

5 The five-layer structure

Motion in the HCSM is a composition of five layers. Each layer has a state space and an operator; the transitions are fixed by dimensional analysis and by the boundary functional.

Layer 1: the substrate. State space $\mathcal{K}_{14} \subset \mathbb{C}^{64}$. Operator: the winding generator $H_{\text{wind}} = i \sin \phi$ (Section 3).

Layer 2: the dimensional measurement equilibrium (DME). State space: the dimensional deviation $\epsilon = D - 4$. Operator: the entropy well $S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4)$, with γ from HCSM-54. The DME has no internal dynamics.

Layer 3: the Core. State space: the Core amplitude $\epsilon(\lambda)$. Operator: the gradient flow $d\epsilon/d\lambda = -2\gamma\epsilon$.

Layer 4: the EFT. State space: the EFT scalar perturbation $\phi(x, t)$. Operator: the mode equation $\ddot{\phi} + \Gamma_{\text{eff}}\dot{\phi} + (k^2/a^2 + m_{\text{eff}}^2)\phi = 0$ with $\Gamma_{\text{eff}} = (3 + \zeta H/\rho)H_{\text{local}}$ and $m_{\text{eff}}^2 = m_s^2 + \zeta R$.

Layer 5: the processor T . State space: $\mathbb{C}^{14}$. Operator: the rank-10 projection $T = P_{\text{mesh}} + P_{\tau}$.

5.1 The stability coefficient γ

The DME stability coefficient is (HCSM-54)

$$\gamma = \frac{c_2}{I(2)} = 0.70283946799007245929257819650854315216110316215188 \dots \quad (15)$$

where $c_2 = 2I''(2)$ and $I(2) = 1 - \text{AGM}(1, \sqrt{2})/2$.

The EFT parameters are (HCSM-54, Corollary 6.2)

$$w = -1 - \frac{(1 - \gamma)^2}{2\pi} = -1.0140540788576648688 \dots, \quad (16)$$

$$\frac{\zeta H}{\rho} = \frac{(1 - \gamma)^2}{6\pi} = 0.004684692952554956 \dots \quad (17)$$

The Core relaxation time is $T_{\text{DME}} = 1/(2\gamma) = 0.711400003516965862 \dots$, and the substrate-DME clock mismatch is $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2 = 0.012143066491921761 \dots$

6 The transition rule

6.1 Transition 1 $\rightarrow$ 2: substrate to DME

Theorem 6.1 (Transition rule). *At the substrate-to-DME boundary, the unique dimensionless quantity formed from the standing-wave frequency ω and the DME relaxation time $T_{\text{DME}} = 1/(2\gamma)$ is*

$$\Phi_{1 \rightarrow 2} = \omega T_{\text{DME}}. \quad (18)$$

Proof. The only available quantities are ω (units 1/time) and T_{DME} (units time). Their product is the unique dimensionless combination. $\square$

With $\omega = 2\sqrt{2}$ for edge tiers and $\omega = 6\sqrt{2}$ for the middle tier:

$$\Phi_{1 \rightarrow 2}^{\text{edge}} = 2 + \delta_{\text{edge}}, \quad \Phi_{1 \rightarrow 2}^{\text{mid}} = 6 + 3\delta_{\text{edge}}. \quad (19)$$

The residual δ_{edge} scales by 3 for the middle tier.

6.2 Transition 2 $\rightarrow$ 3: DME to Core

Definition 6.2 (Boundary functional). *Define the DME-Core interface functional*

$$J[\epsilon_0] = \epsilon_0 + \Lambda \cdot \mathcal{W}[\epsilon_0 \leq |w + 1|], \quad (20)$$

where $\Lambda \rightarrow \infty$ enforces the constraint $\epsilon_0 > |w + 1|$.

Theorem 6.3 (Allowed values). *The unique ϵ_0 of the form $n\delta_{\text{edge}}$ ($n \in \mathbb{Z}_{>0}$) minimizing J is*

$$\epsilon_0 = \lceil |w + 1|/\delta_{\text{edge}} \rceil \cdot \delta_{\text{edge}}. \quad (21)$$

Proof. J is finite only if $\epsilon_0 > |w + 1|$. Among multiples of δ_{edge} , the smallest satisfying this is $n^*\delta_{\text{edge}}$ with $n^* = \lceil |w + 1|/\delta_{\text{edge}} \rceil$. $\square$

Corollary 6.4. *With $|w + 1|/\delta_{\text{edge}} = 1.1573747757\dots$, $\lceil \cdot \rceil = 2$. Therefore the minimum allowed ϵ_0 is $2\delta_{\text{edge}}$ for edge tiers.*

Theorem 6.5 (Cycle-quantized rule). *The DME cycle count for tier τ is $n_\tau = \lfloor \omega_\tau T_{\text{DME}} \rfloor$. Since $n_\tau \geq 2$ for all tiers, the boundary constraint is satisfied automatically, and the initial perturbation is*

$$\epsilon_0(\tau) = n_\tau \cdot \delta_{\text{edge}}. \quad (22)$$

Proof. The DME phase accumulated over one tier period is $\Phi_\tau = \omega_\tau T_{\text{DME}} = n_\tau + \text{frac}$, where $n_\tau = \lfloor \Phi_\tau \rfloor$. The integer n_τ counts complete DME cycles per standing-wave period. The per-cycle residual phase is δ_{edge} . The total residual perturbation carried forward is $n_\tau \delta_{\text{edge}}$. For edge tiers $n_\tau = 2$; for the middle tier $n_\tau = 6$. Both satisfy $n_\tau \delta_{\text{edge}} > |w + 1|$. $\square$

Remark 6.6. *The floor function arises as the integer quantization of DME cycles across the boundary. It is not fitted: it is the unique integer count consistent with the constraint $\epsilon_0 > |w + 1|$ and the physical DME cycle accumulation.*

7 The Core and EFT layers

7.1 Core dynamics

Definition 7.1 (Core dynamics). $d\epsilon/d\lambda = -2\gamma\epsilon$.

Theorem 7.2 (Solution). $\epsilon(\lambda) = \epsilon_0 e^{-2\gamma\lambda}$ with $\tau_{\text{relax}} = 1/(2\gamma) = 0.711400003516965862\dots$

Theorem 7.3 (Lyapunov functional). $L[\epsilon] = \gamma\epsilon^2$ satisfies $dL/d\lambda = -4\gamma^2\epsilon^2 \leq 0$, strict for $\epsilon \neq 0$.

Theorem 7.4 (No oscillation). *The characteristic equation $s + 2\gamma = 0$ has real root $s = -2\gamma$. There are no oscillatory modes.*

7.2 EFT layer

Definition 7.5 (EFT parameters).

$$w = -1 - \frac{(1 - \gamma)^2}{2\pi} = -1.0140540788576648688\dots, \quad (23)$$

$$\frac{\zeta H}{\rho} = \frac{(1 - \gamma)^2}{6\pi} = 0.004684692952554956\dots, \quad (24)$$

$$m_s = \frac{M_P}{N} \exp\left(-\frac{(d+1)N}{d-1}\right) = 9.0247487584 \times 10^{-30} \text{ GeV}. \quad (25)$$

Theorem 7.6 (Mode equation). *A scalar perturbation $\phi(x, t)$ satisfies*

$$\ddot{\phi} + \Gamma_{\text{eff}} \dot{\phi} + \left(\frac{k^2}{a^2} + m_{\text{eff}}^2\right) \phi = 0, \quad (26)$$

with $\Gamma_{\text{eff}} = (3 + \zeta H/\rho)H_{\text{local}}$ and $m_{\text{eff}}^2 = m_s^2 + \zeta R$.

Theorem 7.7 (EFT regime). *For the framework values,*

$$\frac{\zeta R}{m_s^2} = 5.9 \times 10^{-26}, \quad \frac{\Gamma_{\text{eff}}}{\omega_0} = 5.2 \times 10^{-13}, \quad \frac{k}{m_s} < 2.7 \times 10^{-13}. \quad (27)$$

The EFT is extremely underdamped, mass-dominated, and gradient-negligible.

Corollary 7.8. *All 14 modes oscillate at the same frequency $\omega = m_s$ and decay at the same rate $\Gamma_{\text{eff}}/2$.*

8 The processor T

Definition 8.1 (Processor). $T = P_{\text{mesh}} + P_\tau$, where P_{mesh} projects onto $\{\nu, e, u, d, s, \mu, c, b, t\}$ and P_τ onto τ .

Theorem 8.2 (Processor properties). T is an orthogonal projection: $T^2 = T$, $T^\dagger = T$, $\text{rank}(T) = 10$. Its kernel is $\{\text{dark}, W, Z, H\}$.

Theorem 8.3 (Charge bookkeeping).

$$\sum_{j \in \text{im } T} Q_j = -2, \quad \sum_{j \in \ker T} Q_j = +1, \quad \sum_{j=1}^{14} Q_j = -1. \quad (28)$$

9 The loop operator

Definition 9.1 (Loop operator). $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$.

Theorem 9.2 (Closed-form eigenvalue magnitudes). *For each surviving mode m ,*

$$|M_{\text{loop}}(m)| = \frac{|w+1|}{|\omega_m T_{\text{DME}}| \delta_{\text{edge}}} \cdot \exp\left(-\frac{\zeta H}{\rho} \cdot \frac{T_{\text{beat}}}{2}\right). \quad (29)$$

For $m \in \ker T$, $M_{\text{loop}}(m) = 0$.

Proof. $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$ with

$$\begin{aligned} |L_{\text{FDLM}}(m)| &= 1, \\ |L_{\text{DME}}(m)| &= 1, \\ |L_{\text{Core}}(m)| &= e^{-2\gamma\lambda_{\text{EFT}}(m)} = \frac{|w+1|}{\epsilon_0(m)} = \frac{|w+1|}{|\omega_m T_{\text{DME}}| \delta_{\text{edge}}}, \\ |L_{\text{EFT}}(m)| &= e^{-(\zeta H/\rho)T_{\text{beat}}/2}. \end{aligned}$$

Multiplying gives the result. $\square$

Corollary 9.3 (Exact forms).

$$|M_A| = \frac{|w+1|}{2\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2}, \quad (30)$$

$$|M_B| = \frac{|w+1|}{6\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2}. \quad (31)$$

Theorem 9.4 (Exact ratio). $|M_A|/|M_B| = 3 = d - 1$.

Proof. The ratio is $6/2 = 3$. All constants cancel. $\square$

Corollary 9.5 (Substituting framework constants).

$$\frac{|w+1|}{\delta_{\text{edge}}} = \frac{(1-\gamma)^2/(2\pi)}{\sqrt{2}/\gamma - 2}, \quad e^{-(\zeta H/\rho)T_{\text{beat}}/2} = \exp\left(-\frac{(1-\gamma)^2}{6\sqrt{2}}\right). \quad (32)$$

Hence

$$|M_A| = \frac{(1-\gamma)^2}{4\pi\delta_{\text{edge}}} \exp\left(-\frac{(1-\gamma)^2}{6\sqrt{2}}\right), \quad (33)$$

$$|M_B| = \frac{(1-\gamma)^2}{12\pi\delta_{\text{edge}}} \exp\left(-\frac{(1-\gamma)^2}{6\sqrt{2}}\right). \quad (34)$$

Theorem 9.6 (Dissipativity). *Every eigenvalue satisfies $|M_j| < 1$. Under iteration, $M_{\text{loop}}^k \rightarrow 0$.*

Proof. $|w+1|/\delta_{\text{edge}} = 1.1574\dots$ and $e^{-(\zeta H/\rho)T_{\text{beat}}/2} < 1$. For $n = 2$, $|M_A| = 0.5757\dots$; for $n = 6$, $|M_B| = 0.1919\dots$; both < 1 . The kernel modes have eigenvalue 0. $\square$

10 The Motion Theorem

Theorem 10.1 (Motion Theorem). *Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the HCSM substrate with the 14-mode multiplet $\mathcal{K}_{14}$ at eigenvalue $\lambda = 4$. Let $H_{\text{wind}} = i \sin((\pi/8)M)$ be the winding generator. Then:*

1. H_{wind} is the unique Hermitian generator on $\mathbb{C}^{14}$ determined by the substrate and the spectral constraint $\text{spec}(H_{\text{wind}}) = \{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$. Its characteristic polynomial is $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$.
2. The standing-wave amplitudes are exactly the rationals $28/9, 7/18, 21/16$ for supports $64, 48, 32$, derived from the complex Fourier coefficients $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$.

3. The five layers (substrate, DME, Core, EFT, processor) have well-defined state spaces and unique operators.
4. The transition $1 \rightarrow 2$ is $\Phi = \omega T_{\text{DME}}$; the transition $2 \rightarrow 3$ is $\epsilon_0 = \lfloor \omega T_{\text{DME}} \rfloor \cdot \delta_{\text{edge}}$, derived from the boundary functional $J[\epsilon_0]$.
5. The loop operator $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$ is diagonal, with closed-form eigenvalue magnitudes

$$|M_A| = \frac{|w+1|}{2\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2}, \quad |M_B| = \frac{|w+1|}{6\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2}.$$

6. $|M_A|/|M_B| = 3$ exactly.
7. The charge bookkeeping is $\sum Q_{\text{im}} = -2$, $\sum Q_{\text{ker}} = +1$, total -1 .
8. The loop is dissipative: $|M_j| < 1$ for all j .

11 Derivation status

11.1 Derived from the substrate

1. The winding generator $H_{\text{wind}} = i \sin \phi$.
2. The characteristic polynomial $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$.
3. The spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$.
4. The standing-wave amplitudes $28/9, 7/18, 21/16$.
5. The zero drift $\langle x \rangle_j(T_{\text{beat}}) = \langle x \rangle_j(0)$.
6. The uniqueness of M and of $\sin$.
7. The boundary functional $J[\epsilon_0]$.
8. The floor-function cycle quantization $n_\tau = \lfloor \omega_\tau T_{\text{DME}} \rfloor$.
9. The five-layer structure.
10. The loop operator M_{loop} .
11. The closed-form eigenvalue magnitudes $|M_A|, |M_B|$.
12. The exact ratio $|M_A|/|M_B| = 3$.
13. The dissipativity $|M_j| < 1$.
14. The charge bookkeeping $-2, +1, -1$.

11.2 Disclosed inputs

1. The identification of the 14 modes with one Standard Model generation (HCSM-23).
2. The zero-state vector s (HCSM-22).
3. The stability coefficient γ (HCSM-54).
4. The EFT parameters w and $\zeta H/\rho$ (HCSM-54, Corollary 6.2).

11.3 Open

1. The physical mechanism by which the Wick-rotation rate is fixed.
2. The sub-leading corrections to the standing-wave amplitudes at finite lattice spacing.
3. The interpretation of the loop's phase advance as a physical time unit.

12 Falsifiable predictions

1. **Dissipativity.** Every loop eigenvalue satisfies $|M_j| < 1$; no mode grows under iteration. A detection of a non-dissipative mode in the 14-mode multiplet would falsify Theorem 9.6.
2. **Exact ratio.** $|M_A|/|M_B| = 3$ exactly. A measured deviation from this ratio in the framework's downstream predictions would falsify Theorem 9.4.
3. **Standing-wave amplitudes.** The amplitudes 28/9, 7/18, 21/16 are exact rationals. A deviation in the position-space trajectory of any mode would falsify Theorem 3.8.
4. **Floor-function cycle count.** The DME cycle counts are $n_{\text{edge}} = 2$ and $n_{\text{mid}} = 6$. A different count would falsify Theorem 6.5.

13 Conclusion

We have derived the motion of particles in the Hodge Complex Standard Model from first principles. Motion is a five-layer loop with well-defined state spaces and unique operators. Every quantity in the loop is derived analytically: the characteristic polynomial of H_{wind} , the standing-wave amplitudes 28/9, 7/18, 21/16, the boundary functional $J[\epsilon_0]$, and the loop eigenvalue magnitudes.

The framework now provides first-principles derivations of: the substrate, the gauge structure, the generation count, the cosmological cancellation, the warp factor, the Higgs mass, the Weinberg angle, the kinetic mixing, the CKM and PMNS matrices, the mass map, the vacuum stability coefficient γ , and motion.

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

References

- [1] S. Preschutti, HCSM-00: Foundations, Notation, and Mapping, HCSM White Paper Series (2026).
- [2] S. Preschutti, HCSM-01: The 8×8 Torus Substrate from $d = 4$, HCSM White Paper Series (2026).
- [3] S. Preschutti, HCSM-02: The Hodge Complex on Λ , HCSM White Paper Series (2026).
- [4] S. Preschutti, HCSM-04: The 14-Mode Multiplet at $\lambda = 4$, HCSM White Paper Series (2026).
- [5] S. Preschutti, HCSM-05: Orbit Decomposition and Inverse Participation, HCSM White Paper Series (2026).
- [6] S. Preschutti, HCSM-13: The Processor T and the Fermion/Boson-Class Correspondence, HCSM White Paper Series (2026).
- [7] S. Preschutti, HCSM-14: The Winding Generator Spectrum and Class Partition, HCSM White Paper Series (2026).
- [8] S. Preschutti, HCSM-22: The Zero State, HCSM White Paper Series (2026).
- [9] S. Preschutti, HCSM-23: The Identification of the 14 Modes with One Standard Model Generation, HCSM White Paper Series (2026).
- [10] S. Preschutti, HCSM-24: The Charge Theorem, HCSM White Paper Series (2026).
- [11] S. Preschutti, HCSM-54: Vacuum Stability and the Stability Coefficient, HCSM White Paper Series (2026).

A Reproducible verification

The verification script `12-VERIFICATION/HCSM-20_verification.py` verifies:

1. The 14-mode multiplet at $\lambda = 4$ and its D_4 orbits.
2. The winding generator $H_{\text{wind}} = i \sin \phi$.
3. The characteristic polynomial $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$.
4. The spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$.
5. The trace identities $\text{tr}(H_{\text{wind}}^2) = 80$, $\text{tr}(H_{\text{wind}}^4) = 2176$.
6. The standing-wave amplitudes $28/9$, $7/18$, $21/16$.
7. The zero drift $\langle x \rangle_j(T_{\text{beat}}) = \langle x \rangle_j(0)$.
8. The floor-function cycle counts $n_{\text{edge}} = 2$, $n_{\text{mid}} = 6$.
9. The loop eigenvalue magnitudes $|M_A| = 0.5757\dots$, $|M_B| = 0.1919\dots$.
10. The exact ratio $|M_A|/|M_B| = 3$.
11. The charge bookkeeping $-2, +1, -1$.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

B Web Crawler Metadata

B.1 Dublin Core

DC.title: HCSM-20: The Motion Theorem
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09
DC.type: Text / Preprint / White Paper
DC.format: application/pdf; text/x-latex
DC.language: en
DC.publisher: Entropia Research Institute
DC.rights: All rights reserved
DC.subject: Hodge Complex Standard Model; HCSM; motion theorem;
winding generator; standing-wave amplitudes; five-layer loop; loop
operator; dissipativity; 8x8 torus
DC.description: We derive the motion of particles in the Hodge Complex
Standard Model from the discrete substrate. Motion is a five-layer
loop with well-defined state spaces and unique operators. The winding
generator $H_{\text{wind}} = i \sin((\pi/8)M)$ has characteristic polynomial
 $\lambda^{10}(\lambda^2-8)(\lambda^2-32)$. The standing-wave amplitudes are
exactly 28/9, 7/18, 21/16 for the three D4 orbits. The loop eigenvalue
magnitudes satisfy $|M_A|/|M_B| = 3$ exactly. Every number is computed
analytically; no parameter is fitted.
DC.identifier: HCSM-20-2026-09

B.2 BibTeX for Automated Indexing

```
@article{PreschuttiHCSM20,  
  author = {Preschutti, Stanley},  
  title = {{HCSM-20}: The Motion Theorem},  
  journal = {HCSM White Paper Series},  
  year = {2026},  
  month = {September},  
  note = {ORCID: 0009-0004-5445-1744},  
  keywords = {Hodge Complex Standard Model, HCSM, motion theorem,  
    winding generator, standing-wave amplitudes, five-layer loop, loop  
    operator, dissipativity, 8x8 torus},  
  abstract = {We derive the motion of particles in the Hodge Complex  
    Standard Model from the discrete substrate. Motion is a five-layer  
    loop with well-defined state spaces and unique operators. The  
    winding generator  $H_{\text{wind}} = i \sin((\pi/8)M)$  has characteristic  
    polynomial  $\lambda^{10}(\lambda^2-8)(\lambda^2-32)$ . The standing-wave  
    amplitudes are exactly 28/9, 7/18, 21/16. The loop eigenvalue  
    magnitudes satisfy  $|M_A|/|M_B| = 3$  exactly.}
```

}